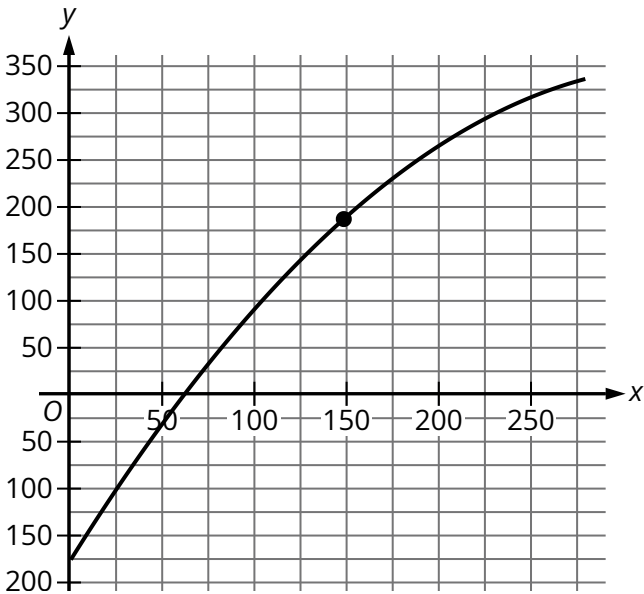


| Assessment | Test | Domain        | Skill               | Difficulty                                          |
|------------|------|---------------|---------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Medium <div><div></div><div></div><div></div></div> |

Question



The graph shows the estimated boiling point  $y$ , in degrees Celsius, of a normal paraffin with a molecular weight of  $x$  grams per mole, where  $1 \leq x \leq 280$ . Which statement is the best interpretation of the point  $(149.02, 186.05)$ ?

Answer

- A. A normal paraffin with a molecular weight of **186.05** grams per mole has an estimated boiling point of **149.02** degrees Celsius.
- B. A normal paraffin with a molecular weight of **149.02** grams per mole has an estimated boiling point of **186.05** degrees Celsius.
- C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of **149.02** grams per mole and an estimated boiling point of **186.05** degrees Celsius.
- D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of **149.02** grams per mole and an estimated boiling point of **186.05** degrees Celsius.

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph shows the estimated boiling point  $y$ , in degrees Celsius, of a normal paraffin with a molecular weight of  $x$  grams per mole. It follows that for the point  $(149.02, 186.05)$ , **149.02** represents the molecular weight, in grams per mole, of a normal paraffin and **186.05** represents its estimated boiling point, in degrees Celsius. Therefore, the best interpretation of the point  $(149.02, 186.05)$  is that a normal paraffin with a molecular weight of **149.02** grams per mole has an estimated boiling point of **186.05** degrees Celsius.

Choice A is incorrect. This statement represents a situation where  $x$ , not  $y$ , represents the boiling point, in degrees Celsius, and  $y$ , not  $x$ , represents the molecular weight, in grams, per mole.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.